If most of your checks were:

...confidently. Congratulations! You have achieved the objectives in this section. Reflect on the study skills you used so that you can continue to use them. What did you do to become confident of your ability to do these things? Be specific.

...with some help. This must be addressed quickly because topics you do not master become potholes in your road to success. In math, every topic builds upon previous work. It is important to make sure you have a strong foundation before you move on. Whom can you ask for help? Your fellow classmates and instructor are good resources. Is there a place on campus where math tutors are available? Can your study skills be improved?

...no—I don't get it! This is a warning sign and you must not ignore it. You should get help right away or you will quickly be overwhelmed. See your instructor as soon as you can to discuss your situation. Together you can come up with a plan to get you the help you need.

7.2 Commutative and Associative Properties

Learning Objectives

By the end of this section, you will be able to:

- Use the commutative and associative properties
- Evaluate expressions using the commutative and associative properties
- Simplify expressions using the commutative and associative properties

☑ **BE PREPARED** 7.4 Before you get started, take this readiness quiz.

Simplify: $7y + 2 + y + 13$.
If you missed this problem, review Example 2.22.

☑ **BE PREPARED** 7.5 Multiply: $\frac{2}{3} \cdot 18$.
If you missed this problem, review Example 4.28.

☑ **BE PREPARED** 7.6 Find the opposite of 15.
If you missed this problem, review Example 3.3.

In the next few sections, we will take a look at the properties of real numbers. Many of these properties will describe things you already know, but it will help to give names to the properties and define them formally. This way we'll be able to refer to them and use them as we solve equations in the next chapter.

Use the Commutative and Associative Properties

Think about adding two numbers, such as 5 and 3.

$5 + 3$	$3 + 5$
8	8

The results are the same. $5 + 3 = 3 + 5$

Notice, the order in which we add does not matter. The same is true when multiplying 5 and 3.

$5 \cdot 3$	$3 \cdot 5$
15	15

Again, the results are the same! $5 \cdot 3 = 3 \cdot 5$. The order in which we multiply does not matter.

These examples illustrate the commutative properties of addition and multiplication.

Commutative Properties

Commutative Property of Addition: if a and b are real numbers, then

$$a + b = b + a$$

Commutative Property of Multiplication: if a and b are real numbers, then

$$a \cdot b = b \cdot a$$

The commutative properties have to do with order. If you change the order of the numbers when adding or multiplying, the result is the same.

EXAMPLE 7.5

Use the commutative properties to rewrite the following expressions:

ⓐ $-1 + 3 =$_____ ⓑ $4 \cdot 9 =$_____

Solution

ⓐ

	$-1 + 3 =$_____
Use the commutative property of addition to change the order.	$-1 + 3 = 3 + (-1)$

ⓑ

	$4 \cdot 9 =$_____
Use the commutative property of multiplication to change the order.	$4 \cdot 9 = 9 \cdot 4$

TRY IT 7.9 Use the commutative properties to rewrite the following:

ⓐ $-4 + 7 =$_____ ⓑ $6 \cdot 12 =$_____

TRY IT 7.10 Use the commutative properties to rewrite the following:

ⓐ $14 + (-2) =$_____ ⓑ $3(-5) =$_____

What about subtraction? Does order matter when we subtract numbers? Does $7 - 3$ give the same result as $3 - 7$?

$$\begin{array}{cc} 7 - 3 & 3 - 7 \\ 4 & -4 \end{array}$$

$$4 \neq -4$$

The results are not the same. $7 - 3 \neq 3 - 7$

Since changing the order of the subtraction did not give the same result, we can say that subtraction is not commutative.

Let's see what happens when we divide two numbers. Is division commutative?

$$\begin{array}{cc} 12 \div 4 & 4 \div 12 \\ \frac{12}{4} & \frac{4}{12} \\ 3 & \frac{1}{3} \end{array}$$

$$3 \neq \frac{1}{3}$$

The results are not the same. So $12 \div 4 \neq 4 \div 12$

Since changing the order of the division did not give the same result, division is not commutative.

Addition and multiplication are commutative. Subtraction and division are not commutative.

Suppose you were asked to simplify this expression.

$$7 + 8 + 2$$

How would you do it and what would your answer be?

Some people would think $7 + 8$ is 15 and then $15 + 2$ is 17. Others might start with $8 + 2$ makes 10 and then $7 + 10$ makes 17.

Both ways give the same result, as shown in Figure 7.3. (Remember that parentheses are grouping symbols that indicate which operations should be done first.)

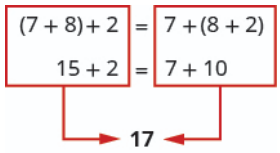


Figure 7.3

When adding three numbers, changing the grouping of the numbers does not change the result. This is known as the Associative Property of Addition.

The same principle holds true for multiplication as well. Suppose we want to find the value of the following expression:

$$5 \cdot \frac{1}{3} \cdot 3$$

Changing the grouping of the numbers gives the same result, as shown in Figure 7.4.

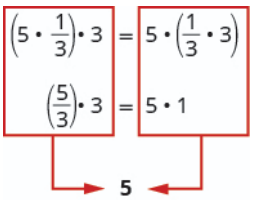


Figure 7.4

When multiplying three numbers, changing the grouping of the numbers does not change the result. This is known as the Associative Property of Multiplication.

If we multiply three numbers, changing the grouping does not affect the product.

You probably know this, but the terminology may be new to you. These examples illustrate the *Associative Properties*.

Associative Properties

Associative Property of Addition: if a, b, and c are real numbers, then

$$(a + b) + c = a + (b + c)$$

Associative Property of Multiplication: if a, b, and c are real numbers, then

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

EXAMPLE 7.6

Use the associative properties to rewrite the following:

ⓐ $(3 + 0.6) + 0.4 =$_____ ⓑ $\left(-4 \cdot \frac{2}{5}\right) \cdot 15 =$_____

Solution

ⓐ

	$(3+0.6)+0.4=$_____
Change the grouping.	$(3+0.6)+0.4=3+(0.6+0.4)$

Notice that $0.6+0.4$ is 1, so the addition will be easier if we group as shown on the right.

ⓑ

	$\left(-4\cdot\frac{2}{5}\right)\cdot 15=$_____
Change the grouping.	$\left(-4\cdot\frac{2}{5}\right)\cdot 15=-4\cdot\left(\frac{2}{5}\cdot 15\right)$

Notice that $\frac{2}{5}\cdot 15$ is 6. The multiplication will be easier if we group as shown on the right.

TRY IT 7.11 Use the associative properties to rewrite the following:
ⓐ $(1+0.7)+0.3=$_____ ⓑ $(-9\cdot 8)\cdot\frac{3}{4}=$_____

TRY IT 7.12 Use the associative properties to rewrite the following:
ⓐ $(4+0.6)+0.4=$_____ ⓑ $(-2\cdot 12)\cdot\frac{5}{6}=$_____

Besides using the associative properties to make calculations easier, we will often use it to simplify expressions with variables.

EXAMPLE 7.7

Use the Associative Property of Multiplication to simplify: $6\,(3x)$.

Solution

	$6\,(3x)$
Change the grouping.	$(6\cdot 3)\,x$
Multiply in the parentheses.	$18x$

Notice that we can multiply $6\cdot 3$, but we could not multiply $3\cdot x$ without having a value for x.

TRY IT 7.13 Use the Associative Property of Multiplication to simplify the given expression: $8\,(4x)$.

TRY IT 7.14 Use the Associative Property of Multiplication to simplify the given expression: $-9\,(7y)$.

Evaluate Expressions using the Commutative and Associative Properties

The commutative and associative properties can make it easier to evaluate some algebraic expressions. Since order does not matter when adding or multiplying three or more terms, we can rearrange and re-group terms to make our work easier, as the next several examples illustrate.

EXAMPLE 7.8

Evaluate each expression when $x = \frac{7}{8}$.

ⓐ $x + 0.37 + (-x)$ ⓑ $x + (-x) + 0.37$

Solution

ⓐ

	$x + 0.37 + (-x)$
Substitute $\frac{7}{8}$ for x.	$\frac{7}{8} + 0.37 + \left(-\frac{7}{8}\right)$
Convert fractions to decimals.	$0.875 + 0.37 + (-0.875)$
Add left to right.	$1.245 - 0.875$
Subtract.	0.37

ⓑ

	$x + (-x) + 0.37$
Substitute $\frac{7}{8}$ for x.	$\frac{7}{8} + \left(-\frac{7}{8}\right) + 0.37$
Add opposites first.	0.37

What was the difference between part ⓐ and part ⓑ ? Only the order changed. By the Commutative Property of Addition, $x + 0.37 + (-x) = x + (-x) + 0.37$. But wasn't part ⓑ much easier?

TRY IT 7.15 Evaluate each expression when $y = \frac{3}{8}$: ⓐ $y + 0.84 + (-y)$ ⓑ $y + (-y) + 0.84$.

TRY IT 7.16 Evaluate each expression when $f = \frac{17}{20}$: ⓐ $f + 0.975 + (-f)$ ⓑ $f + (-f) + 0.975$.

Let's do one more, this time with multiplication.

EXAMPLE 7.9

Evaluate each expression when $n = 17$.

ⓐ $\frac{4}{3}\left(\frac{3}{4}n\right)$ ⓑ $\left(\frac{4}{3} \cdot \frac{3}{4}\right) n$

Solution

ⓐ

	$\frac{4}{3}\left(\frac{3}{4}n\right)$

Substitute 17 for n.	$\frac{4}{3}\left(\frac{3}{4}\cdot 17\right)$
Multiply in the parentheses first.	$\frac{4}{3}\left(\frac{51}{4}\right)$
Multiply again.	17

ⓑ

	$\left(\frac{4}{3}\cdot\frac{3}{4}\right)n$
Substitute 17 for n.	$\left(\frac{4}{3}\cdot\frac{3}{4}\right)\cdot 17$
Multiply. The product of reciprocals is 1.	$(1)\cdot 17$
Multiply again.	17

What was the difference between part ⓐ and part ⓑ here? Only the grouping changed. By the Associative Property of Multiplication, $\frac{4}{3}\left(\frac{3}{4}n\right)=\left(\frac{4}{3}\cdot\frac{3}{4}\right)n$. By carefully choosing how to group the factors, we can make the work easier.

TRY IT 7.17 Evaluate each expression when $p=24$: ⓐ $\frac{5}{9}\left(\frac{9}{5}p\right)$ ⓑ $\left(\frac{5}{9}\cdot\frac{9}{5}\right)p$.

TRY IT 7.18 Evaluate each expression when $q=15$: ⓐ $\frac{7}{11}\left(\frac{11}{7}q\right)$ ⓑ $\left(\frac{7}{11}\cdot\frac{11}{7}\right)q$

Simplify Expressions Using the Commutative and Associative Properties

When we have to simplify algebraic expressions, we can often make the work easier by applying the Commutative or Associative Property first instead of automatically following the order of operations. Notice that in Example 7.8 part ⓑ was easier to simplify than part ⓐ because the opposites were next to each other and their sum is 0. Likewise, part ⓑ in Example 7.9 was easier, with the reciprocals grouped together, because their product is 1. In the next few examples, we'll use our number sense to look for ways to apply these properties to make our work easier.

EXAMPLE 7.10

Simplify: $-84n+(-73n)+84n$.

Solution

Notice the first and third terms are opposites, so we can use the commutative property of addition to reorder the terms.

	$-84n+(-73n)+84n$
Re-order the terms.	$-84n+84n+(-73n)$
Add left to right.	$0+(-73n)$
Add.	$-73n$

TRY IT 7.19 Simplify: $-27a + (-48a) + 27a$.

TRY IT 7.20 Simplify: $39x + (-92x) + (-39x)$.

Now we will see how recognizing reciprocals is helpful. Before multiplying left to right, look for reciprocals—their product is 1.

EXAMPLE 7.11

Simplify: $\frac{7}{15} \cdot \frac{8}{23} \cdot \frac{15}{7}$.

Solution

Notice the first and third terms are reciprocals, so we can use the Commutative Property of Multiplication to reorder the factors.

	$\frac{7}{15} \cdot \frac{8}{23} \cdot \frac{15}{7}$
Re-order the terms.	$\frac{7}{15} \cdot \frac{15}{7} \cdot \frac{8}{23}$
Multiply left to right.	$1 \cdot \frac{8}{23}$
Multiply.	$\frac{8}{23}$

TRY IT 7.21 Simplify: $\frac{9}{16} \cdot \frac{5}{49} \cdot \frac{16}{9}$.

TRY IT 7.22 Simplify: $\frac{6}{17} \cdot \frac{11}{25} \cdot \frac{17}{6}$.

In expressions where we need to add or subtract three or more fractions, combine those with a common denominator first.

EXAMPLE 7.12

Simplify: $\left(\frac{5}{13} + \frac{3}{4}\right) + \frac{1}{4}$.

Solution

Notice that the second and third terms have a common denominator, so this work will be easier if we change the grouping.

	$\left(\frac{5}{13} + \frac{3}{4}\right) + \frac{1}{4}$
Group the terms with a common denominator.	$\frac{5}{13} + \left(\frac{3}{4} + \frac{1}{4}\right)$
Add in the parentheses first.	$\frac{5}{13} + \left(\frac{4}{4}\right)$
Simplify the fraction.	$\frac{5}{13} + 1$

Add.	$1\frac{5}{13}$
Convert to an improper fraction.	$\frac{18}{13}$

TRY IT 7.23 Simplify: $\left(\frac{7}{15}+\frac{5}{8}\right)+\frac{3}{8}$.

TRY IT 7.24 Simplify: $\left(\frac{2}{9}+\frac{7}{12}\right)+\frac{5}{12}$.

When adding and subtracting three or more terms involving decimals, look for terms that combine to give whole numbers.

EXAMPLE 7.13

Simplify: $(6.47q+9.99q)+1.01q$.

Solution

Notice that the sum of the second and third coefficients is a whole number.

	$(6.47q+9.99q)+1.01q$
Change the grouping.	$6.47q+(9.99q+1.01q)$
Add in the parentheses first.	$6.47q+(11.00q)$
Add.	$17.47q$

Many people have good number sense when they deal with money. Think about adding 99 cents and 1 cent. Do you see how this applies to adding $9.99+1.01$?

TRY IT 7.25 Simplify: $(5.58c+8.75c)+1.25c$.

TRY IT 7.26 Simplify: $(8.79d+3.55d)+5.45d$.

No matter what you are doing, it is always a good idea to think ahead. When simplifying an expression, think about what your steps will be. The next example will show you how using the Associative Property of Multiplication can make your work easier if you plan ahead.

EXAMPLE 7.14

Simplify the expression: $[1.67(8)](0.25)$.

Solution

Notice that multiplying $(8)(0.25)$ is easier than multiplying $1.67(8)$ because it gives a whole number. (Think about having 8 quarters—that makes $\$2$.)

	$[1.67(8)](0.25)$
Regroup.	$1.67[(8)(0.25)]$

Multiply in the brackets first.	$1.67\,[2]$
Multiply.	3.34

TRY IT 7.27 Simplify: $[1.17(4)]\,(2.25)$.

TRY IT 7.28 Simplify: $[3.52(8)]\,(2.5)$.

When simplifying expressions that contain variables, we can use the commutative and associative properties to re-order or regroup terms, as shown in the next pair of examples.

EXAMPLE 7.15

Simplify: $6\,(9x)$.

Solution

	$6\,(9x)$
Use the associative property of multiplication to re-group.	$(6 \cdot 9)\,x$
Multiply in the parentheses.	$54x$

TRY IT 7.29 Simplify: $8\,(3y)$.

TRY IT 7.30 Simplify: $12\,(5z)$.

In The Language of Algebra, we learned to combine like terms by rearranging an expression so the like terms were together. We simplified the expression $3x + 7 + 4x + 5$ by rewriting it as $3x + 4x + 7 + 5$ and then simplified it to $7x + 12$. We were using the Commutative Property of Addition.

EXAMPLE 7.16

Simplify: $18p + 6q + (-15p) + 5q$.

Solution

Use the Commutative Property of Addition to re-order so that like terms are together.

	$18p + 6q + (-15p) + 5q$
Re-order terms.	$18p + (-15p) + 6q + 5q$
Combine like terms.	$3p + 11q$

TRY IT 7.31 Simplify: $23r + 14s + 9r + (-15s)$.

TRY IT 7.32 Simplify: $37m + 21n + 4m + (-15n)$.

LINKS TO LITERACY

The *Links to Literacy* activity, "Each Orange Had 8 Slices" will provide you with another view of the topics covered in this section.

SECTION 7.2 EXERCISES

Practice Makes Perfect

Use the Commutative and Associative Properties

In the following exercises, use the commutative properties to rewrite the given expression.

20. $8+9=$___

21. $7+6=$___

22. $8(-12)=$___

23. $7(-13)=$___

24. $(-19)(-14)=$___

25. $(-12)(-18)=$___

26. $-11+8=$___

27. $-15+7=$___

28. $x+4=$___

29. $y+1=$___

30. $-2a=$___

31. $-3m=$___

In the following exercises, use the associative properties to rewrite the given expression.

32. $(11+9)+14=$___

33. $(21+14)+9=$___

34. $(12\cdot 5)\cdot 7=$___

35. $(14\cdot 6)\cdot 9=$___

36. $(-7+9)+8=$___

37. $(-2+6)+7=$___

38. $\left(16\cdot\frac{4}{5}\right)\cdot 15=$___

39. $\left(13\cdot\frac{2}{3}\right)\cdot 18=$___

40. $3(4x)=$___

41. $4(7x)=$___

42. $(12+x)+28=$___

43. $(17+y)+33=$___

Evaluate Expressions using the Commutative and Associative Properties

In the following exercises, evaluate each expression for the given value.

44. If $y=\frac{5}{8}$, evaluate:
(a) $y+0.49+(-y)$
(b) $y+(-y)+0.49$

45. If $z=\frac{7}{8}$, evaluate:
(a) $z+0.97+(-z)$
(b) $z+(-z)+0.97$

46. If $c=-\frac{11}{4}$, evaluate:
(a) $c+3.125+(-c)$
(b) $c+(-c)+3.125$

47. If $d=-\frac{9}{4}$, evaluate:
(a) $d+2.375+(-d)$
(b) $d+(-d)+2.375$

48. If $j=11$, evaluate:
(a) $\frac{5}{6}\left(\frac{6}{5}j\right)$ (b) $\left(\frac{5}{6}\cdot\frac{6}{5}\right)j$

49. If $k=21$, evaluate:
(a) $\frac{4}{13}\left(\frac{13}{4}k\right)$
(b) $\left(\frac{4}{13}\cdot\frac{13}{4}\right)k$

50. If $m=-25$, evaluate:
(a) $-\frac{3}{7}\left(\frac{7}{3}m\right)$
(b) $\left(-\frac{3}{7}\cdot\frac{7}{3}\right)m$

51. If $n=-8$, evaluate:
(a) $-\frac{5}{21}\left(\frac{21}{5}n\right)$
(b) $\left(-\frac{5}{21}\cdot\frac{21}{5}\right)n$

Simplify Expressions Using the Commutative and Associative Properties

In the following exercises, simplify.

52. $-45a+15+45a$

53. $9y+23+(-9y)$

54. $\frac{1}{2}+\frac{7}{8}+\left(-\frac{1}{2}\right)$

55. $\frac{2}{5}+\frac{5}{12}+\left(-\frac{2}{5}\right)$

56. $\frac{3}{20}\cdot\frac{49}{11}\cdot\frac{20}{3}$

57. $\frac{13}{18}\cdot\frac{25}{7}\cdot\frac{18}{13}$

58. $\frac{7}{12}\cdot\frac{9}{17}\cdot\frac{24}{7}$

59. $\frac{3}{10}\cdot\frac{13}{23}\cdot\frac{50}{3}$

60. $-24\cdot 7\cdot\frac{3}{8}$

61. $-36\cdot 11\cdot\frac{4}{9}$

62. $\left(\frac{5}{6}+\frac{8}{15}\right)+\frac{7}{15}$

63. $\left(\frac{1}{12}+\frac{4}{9}\right)+\frac{5}{9}$

64. $\frac{5}{13}+\frac{3}{4}+\frac{1}{4}$

65. $\frac{8}{15}+\frac{5}{7}+\frac{2}{7}$

66. $(4.33p+1.09p)+3.91p$